

connected as a 32W, 4-ohm circuit will also serve the purpose.

The total harmonic distortion

(THD) at 20W is around 10 per cent, which is good enough for such a simple and low-cost amplifier. The noise

is not noticeable. The amplifier is versatile and can be adapted to a variety of signal sources and applications. ●

Analogue Wattmeter

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Unlike current or voltage measurement, the measurement of power is somewhat inconvenient as power is the product of voltage (V) and current (I) flowing through a dissipative circuit.

In a linear-moving coil meter, the deflection of the pointer is proportional to the current flowing through the meter. It is also proportional to the duty cycle of the current waveform. The net deflection is proportional to the product of duty cycle and the magnitude of current. Using this principle, the power in a circuit can be measured. The degree of deflection of the pointer is proportional to the product V_I , giving the meter the property of a power meter.

The figure shows the circuit of an analogue wattmeter. This meter makes three connections (A, B and C) with the test circuit. The current through the test circuit is sampled by measuring the voltage drop across a small resistor (R1) such that its introduction does not affect the current in the test circuit significantly. After amplification of this voltage, it is made to generate a constant current, the magnitude of which is proportional to the voltage drop across series resistor R1.

The voltage that drives test load R_L is sampled through an attenuator circuit and fed to a comparator that compares it with a ramp voltage. The higher the supply voltage of the test circuit, the higher the point in the ramp at which the comparator

fires. Thus pulsewidth modulation becomes proportional to the value of the sampled voltage. The output of the comparator along with the logic gates is used to modulate the flow of current through the meter, making its deflection proportional to V_I in the test circuit.

The values shown in the circuit correspond to a meter capable of measuring (5V, 10V, 20V, 50V) x 500 mA, or 2.5W to 25W. By selecting an appropriate value of R1, it is possible to accommodate higher currents and thus higher power.

IC1, which is configured as a non-inverting voltage amplifier, is used to amplify the voltage across R1. IC2(A) generates a constant current ($I_c = V_i /$

